

# UB Math

Worksheets: Answer all questions on pages titled with  
“Exercises!”

# What is math?

- ▶ Is it a science? Is it art?
- ▶ If its a science does that mean math is discovered?
- ▶ If its like art does that mean its created by humans or invented?



Figure: Is math discovered or invented?

# What is a set?

- ▶ A **set** is a collection of things. For example, we could talk about the set of files on my desktop, a set of turntables or a set of cards.

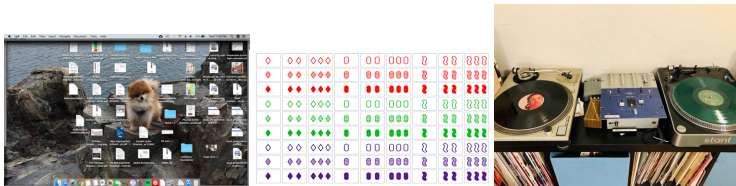


Figure: Examples of sets.

- ▶ To sound a bit more fancy let's call the things inside a set something better than... things.
  - ▶ Let's call them **elements**.
  - ▶ How many elements are in the desktop set?
  - ▶ In the SET GAME, each card has 1 of 3 symbols, 1 of 3 colors, 1 to 3 symbols, and 1 of 3 shadings.
  - ▶ The number of elements in a set has a fancy name: **cardinality**

# What do we do with sets?

- ▶ We can combine sets together or break them apart.
- ▶ There are three special words that help us play with sets: OR, AND, NOT
  - ▶ Let's say these out loud: ♪ OR ♪ AND ♪♪ NOT ♪ 。



Figure: We can eat them? Are they nutritious?

# Combining sets

- ▶ We can create a new set from a couple of sets  $A$  and  $B$  by using the word: **or**.
- ▶ This is called the **union** of  $A$  and  $B$ .
- ▶ We do this by declaring that the union is the set with elements in  $A$  **or**  $B$ .
  - ▶ This word **or** is not very restrictive.
  - ▶ Elements in the union are either in  $A$  or  $B$  (or both).
  - ▶ This means that the union of two sets contains everything from both sets.
  - ▶ All the things!
- ▶ You can take the union of as many sets as you like!
  - ▶ Even infinitely many sets!
  - ▶ Wait, sets can be infinite?
- ▶ We symbolize it as  $A \cup B$ .

# Combining sets



Figure: Sets of snails and shrimps. Union of seafood dishes.

- ▶ You can take the union of as many sets as you like!
  - ▶ Even infinitely many sets!
  - ▶ Wait, sets can be infinite?
- ▶ We symbolize it as  $A \cup B$ .

# Dissecting sets

- ▶ We can create a new set from a couple of sets  $A$  and  $B$  by using the word: **and**.
- ▶ This is called the **intersection** of  $A$  and  $B$  and we write  $A \cap B$ .
- ▶ The intersection is the set with elements in  $A$  **and**  $B$ .
  - ▶ The word **and** here is very strict.
  - ▶ What is an example of the intersection of country music and hip hop?

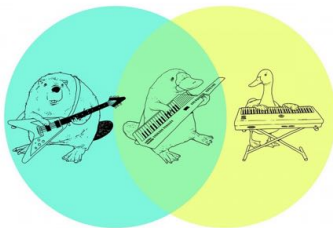


Figure: Venn Diagrams of animals.

# Can you not?

- ▶ We can create a new set from a set  $A$  by using the word: **not**.
- ▶ This is called the **negation** of  $A$  and we write it as  $\neg A$  or  $\sim A$ .
- ▶ This is where things get a little weird.
  - ▶ What does it mean to not be in  $A$ ?
  - ▶ Let  $A$  = set of Tik Toks. What does it mean to not be a Tik Tok?
  - ▶ The set of not Tik Toks is everything in the universe but Tik Toks.
  - ▶ In fact, this set is not a Tik Tok. So it must be in there too.
  - ▶ That means the set of not Tik Toks contains as one of it's elements the set of not Tik Toks.
  - ▶ In other words it contains itself! This can get problematic.

# Can you not? Exercises!

- ▶ I warned you it would get weird... or at least confusing.
- ▶ Because this can be confusing we often set the stage and state what **universe** of elements we intend to discuss.
  - ▶ When talking about sets we give some context by stating a general set called the universe.
  - ▶ So if we talk about Tik Toks our universe set could be social media videos.
  - ▶ Then we write the negation with respect to the universe set as  $T \setminus U$ .
  - ▶ Write three examples.

# Subsets! Exercises!

- ▶ We can also dissect a set by only looking at some of its elements.
- ▶ This is called a **subset**.
- ▶ The magic words for this concept are in the phrase: **If ... then...**
- ▶ The set  $A$  is a subset of  $B$  means the following is true: If an element is in  $A$  **then** it is in  $B$ .
- ▶ We write  $A \subset B$ .
  - ▶ What are some examples of subsets of the set of things in your house?

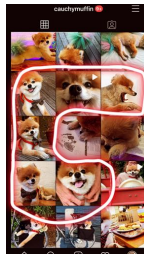


Figure: Subset examples

# Staring into nothingness! Exercises!

- ▶ There is a strange but useful set called the empty set.
- ▶ Yep, that's exactly what it is. It's the set with nothing in it.
- ▶ We write the empty set as  $\emptyset$  or  $\varnothing$  or  $\{\}$ .
  - ▶ What is the cardinality of  $\emptyset$ ?
  - ▶ What is  $\emptyset \cup A$ ?
  - ▶ What is  $\emptyset \cap A$ ?

# Infinite sets!

- ▶ Let's look at a set with only one element  $\{a\}$ . We call this a singleton set.
- ▶ Now look at this weird set  $\{\{a\}\}$ . Describe this set in words. What is its cardinality?
  - ▶ What's the union of these two sets?
  - ▶  $\{a\} \cup \{\{a\}\} = \{a, \{a\}\}$  What is the cardinality of this set?
  - ▶ What is we do another union?  $\{a, \{a\}\} \cup \{\{a, \{a\}\}\}$
  - ▶ In other words, we take a set  $A$  and union it with the set  $\{A\}$ . Then repeat.
  - ▶ When does it stop? We just invented infinity! More precisely we call this process **induction**.

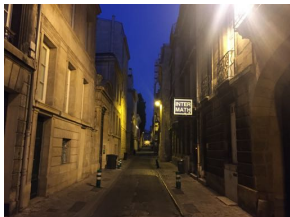


Figure: Infinite math alley: once you enter you never return.

# Unionize!

- ▶ What is a number? Is it how many fingers you hold up? How many followers you have? What is 7? What is it made of?
- ▶ Let's play a union game. Start with the empty set  $\emptyset$ .
  - ▶ Then like we did above, union it with the set that contains it:  $\emptyset \cup \{\{\emptyset\}\}$ . What do we get?
  - ▶ What is the cardinality of this set?
  - ▶ Now let's play this union game on repeat:  $\{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\}$ .
  - ▶ What is the cardinality?
  - ▶ And repeat a few more times for fun:  $\{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\}$ ,  $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \cup \{\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\}$ ...

# Numbers! Exercises!

- ▶ This got ugly quick! Lets look at the cardinality and do some name calling.
  - ▶ Let's define the **natural numbers** by these sets according to their cardinality.
  - ▶ Define the number **zero** as the **empty set**.
  - ▶ Then given a number  $n$ , the next number,  $n + 1$  is constructed by taking union of  $n$  with the set containing  $n$ .
  - ▶  $n + 1 = n \cup \{n\}$ .
  - ▶ Let's compute a few below:

$0 = \emptyset$   
 $1 = \{\emptyset\}$   
 $2 = \{\emptyset, \{\emptyset\}\}$   
 $3 =$   
 $4 =$   
 $5 =$



# Numbers Summary!

- ▶ Each number is a set whose elements are the previous numbers.

- ▶ ex:  $3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} = \{0, 1, 2\}$

- ▶ Each number is a set with that many elements.

- ▶ ex:  $4 = \left\{ \overbrace{\emptyset}^0, \overbrace{\{\emptyset\}}^1, \overbrace{\{\emptyset, \{\emptyset\}\}}^2, \overbrace{\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}}^3 \right\}$

- ▶ Every number is a subset of larger numbers.

- ▶ ex:  $3 \subset 4$  but also  $3 \in 4$  (this means 3 is an element of 4)
  - ▶ This means we can use the idea of subset to talk about order.
  - ▶  $3 \subset 4 \iff 3 < 4$



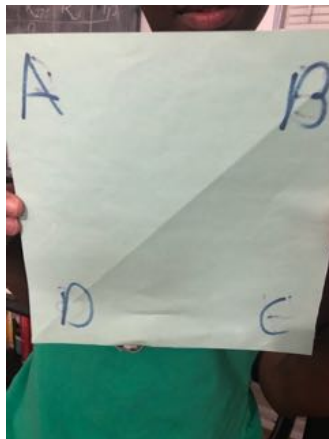
# Square Dance

The rotations and flips of a square act like numbers:

$\{R_0, R_{90}, R_{180}, R_{270}, F_H, F_V, D_L, D_R\}$

They follow similar rules as addition and multiplication of numbers.

- ▶ Combining rotations and flips results in a rotation or flip.
- ▶  $R_0$  is what we call an identity element: it does nothing.
- ▶ Every rotation and flip has cancel buddy.
  - ▶ ex.  $R_{90}R_{270} = R_0$
- ▶ Associative: ex.  
 $(R_{180}F_V)D_R = R_{180}(F_VD_R)$
- ▶ Which elements commute?

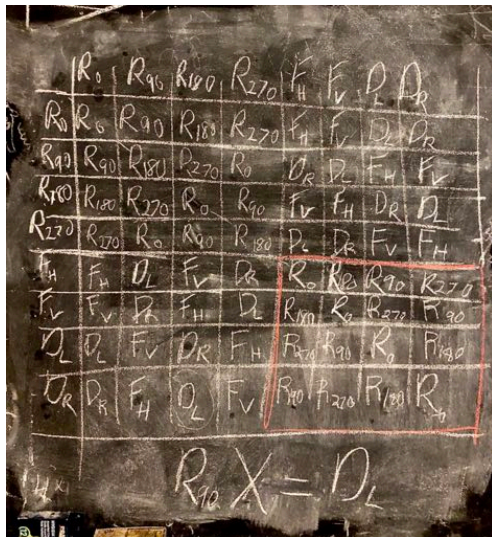


# Square Dance! Exercises!

We can solve equations!

Try these. Solve for  $X$ .

1.  $R_{90}X =$
2.  $D_RX =$
3.  $X^2 = F_H$
4.  $R_{90}X = R_0$
5.  $F_HX = XF_H$
6.  $R_{180}X = XR_{180}$



# Cha Cha Slide! Exercises!

Diagonal left, Diagonal right now y'all...

- ▶ The rules define the dance.
- ▶ The above rules form a fun dance called a *group*.
- ▶ We can play this dance with any polygon: triangle, pentagon, hexagon...
- ▶ What happens if we use a triangle?
- ▶ How many ways can you flip a triangle?
- ▶ What happens if we use a circle?
- ▶ How many ways can you flip a circle?



# 911 is 11, No Joke!

The rotations of a square are similar to the hands on a clock.

- ▶ The clock number system cycles back to 1 after 12.
  - ▶ This means  $12 = 0$  in clock land.
  - ▶ Addition is like telling time: ex.  
 $8 + 7 = 15 = \cancel{12} + 3 = 3$
  - ▶ What are inverses? ex.  $8 + 4 = 12$   
 $\text{mod } 12 = 0$
- ▶ Clockland is a commutative group



# Mod Out!

There are many Clocklands. Instead of 12, it can be any number.

- ▶ This type of number land is called modular arithmetic.
  - ▶ We say  $13 \bmod 7 = 6$
  - ▶ Just divide 13 by 7 and take the remainder.
- ▶ What happens if we multiply?
  - ▶  $4 \times 3 = 12 \bmod 7 = 5$
  - ▶  $4 \times 3 = 12 \bmod 5 = 2$
  - ▶ What are inverses?
- ▶ What are some subgroups?
  - ▶ ex.  $3 + 3 + 3 + 3 = 12 \bmod 4 = 0$
  - ▶ ex.  $2 \times 2 \times 2 \dots = ?$



# Mod Out! Exercises!

Try these:

- ▶  $15 \bmod 7 =$
- ▶  $4 \times 3 = 12 \bmod 9 =$
- ▶  $4 \times 3 = 12 \bmod 2 =$
- ▶  $2 \times 3 = 6 \bmod 5 =$
- ▶  $5 \times 2 = 10 \bmod 5 =$
- ▶  $10 \times 4 = 40 \bmod 3 =$
- ▶  $6 \times 5 = 30 \bmod 14 =$
- ▶  $10 + 4 = 14 \bmod 3 =$
- ▶  $6 + 5 = 11 \bmod 4 =$



# Euclid's Elements

- ▶ In Ancient Greece, 300 BCE, geometry was math and the laws of nature.
  - ▶ Euclid was messenger and messiah.
  - ▶ The Elements: geometry in 13 books
  - ▶ Defines points, lines, angles...
  - ▶ Postulates 5 laws of geometry
  - ▶ The 5 postulates prove theorems
  - ▶ ex. opposite angles are equal
  - ▶ ex. *Interior angles of triangle add to  $180^\circ$*



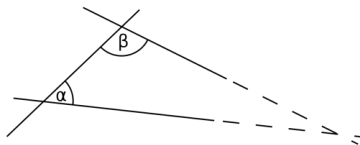
# The Postulates

1. You can draw a straight line between two points.
  - ▶ This seems obvious.
  - ▶ But it implies there are no holes in our space
2. You can extend a straight line as far as you want.
3. You can draw a circle with any center and radius
  - ▶ Again seems obvious but so does the 13th amendment.
4. All right angles are equal
  - ▶ *The equal rights law of geometry*
  - ▶ Seems obvious in geometry



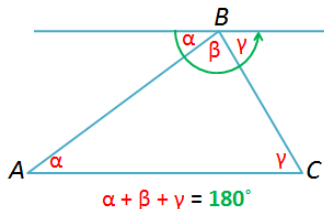
# Euclid Pleads the 5th

- ▶ Get ready. This one is not so simple.
- ▶ 5. If a straight line falling on two straight lines
  - ▶ makes the interior angles on the same side
  - ▶ less than two right angles,
  - ▶ the two straight lines
  - ▶ if drawn far enough,
  - ▶ will meet on that side.
- ▶ In other words, the lines aren't parallel.
  - ▶ *If the interior angles are both right*
  - ▶ Then the lines are parallel.



# Euclid Was a Flat Spacer!

- ▶ The 5th postulate is often called the parallel postulate.
- ▶ It proves that
  - ▶ the interior angles of a triangle is  $180^\circ$ .
  - ▶ Alternate interior angles of  $\parallel$  lines
  - ▶ unique  $\parallel$  line through a point
- ▶ Many tried to prove the 5th
  - ▶ *But all were unsuccessful*
  - ▶ It is a law of its own.
  - ▶ Describes geometry on a flat plane.
- ▶ Geometry class is nothing but Euclid



# Euclid Was a Flat Spacer! Exercises!

- ▶ Assume the horizontal lines in the diagram are parallel.
- ▶ Use the diagram to prove that the sum of the interior angles of a triangle equal  $180^\circ$ .

